



Towards dynamic reference information models: Readiness for ICT mass customisation

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ABSTRACT

Current dynamic demand-driven networks make great demands on, in particular, the interoperability and agility of information systems. This paper investigates how reference information models can be used to meet these demands by enhancing ICT mass customisation. It was found that reference models for Production and Supply Chain Management do not yet sufficiently meet the requirements of a mass customisation approach. They have developed from isolated models based on pure standardisation and tailored customisation strategies, towards static repository-based models founded on segmented standardisation strategies. Existing models provide valuable knowledge for developing towards more dynamic reference information models, including the progress made by ERP vendors to make their reference models configurable. Important remaining challenges are setting up reference information models as generic models that define classes of architectures, and incorporating user-friendly means that guide users through the process of configuring specific information models.

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1. Introduction

In response to today's increasingly volatile business environment, supply chains are in transition from chains pushing products efficiently to the marketplace, towards agile networks that sense and react dynamically to demand information. This makes great demands on, in particular, the interoperability and agility of supporting information systems [1,2]. Interoperability is the ability for two systems to understand one another and to use one another's functionality [3]. ICT agility is the ability to identify needed changes in the information-processing functionalities and to implement them quickly and efficiently [4].

The main challenge in meeting these demands is combining flexible customisation with efficient standardisation. In business literature, mass customisation is broadly advocated as a core approach to balance these seemingly contradictory notions [5,6]. Mass customisation relates the ability to provide customised products or services through flexible processes to the ability to produce in high volumes at reasonable costs [7]. This is by fabricating parts of the product in volume as standard components, while achieving distinctiveness through customer-specific assembly of modules [8].

Recent developments in ICT, particularly Service-Oriented Architecture (SOA), enable the application of mass customisation principles to information systems [9]. ICT mass customisation combines the advantages of standard and customised software. It enables on-demand configuration of information systems from standard components with standardised interfaces.

Information models (e.g. process and data models) support ICT configuration tasks since they provide systematic representations of the complex architectures to be configured. Specific information models could be developed from scratch for each configuration, but this would result in high costs and long lead-times. An alternative approach would be to use available 'best practices' captured in reference information models as a 'frame of reference' (i.e. blueprint, template).

The objective of this paper is to assess how reference information models can be used to enhance ICT mass customisation. Therefore, it respectively aims to: (i) identify the requirements to reference information models in an ICT mass customisation approach, (ii) investigate the extent to which existing reference models, in the domain of Production and Supply Chain Management, are useful for ICT mass customisation, and (iii) to explore trends in the development towards dynamic models.

The research started with a literature study to identify the requirements on reference information models in an ICT mass customisation approach. Therefore, a three-stage funnel approach was followed. First, the generic concept of mass customisation and the requirements on systems, enabling such an approach, were

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defined. Secondly, the approach and identified requirements were applied to mass customisation of information systems. Thirdly, the analysis concentrated on the role of reference information models in mass-customisable ICT.

After literature review, existing reference information models were investigated. Because user acceptance is an important characteristic of reference models, the investigation focused on reference models that are widely acknowledged and applied in the business community. The scope of our analysis was the domain of Production and Supply Chain Management, implying that reference information models which mainly focus on eCommerce processes (ordering, invoicing, catalogue exchange, etc.) and sector-specific reference models were not assessed.

The investigation was carried out through desk study and in-depth expert interviews with reference model developers, implementation consultants and managers (67%, 53% and 33%, respectively, multiple roles possible). In total, 15 experts were interviewed (9 persons from 7 software vendors and 6 people from 5 service providers) in-depth, based on a structured three-part questionnaire: (i) content and technology, (ii) model development, and (iii) implementation and usage.¹

The structure of this paper follows the research approach. Section 2 defines the concept of mass customisation, the requirements on systems enabling such an approach, and applies this to information systems. Section 3 elaborates on the role of reference models and the requirements on such models for the enhancement of ICT mass customisation. Section 4 provides an overview of existing reference models for Production and Supply Chain Management and describes the extent to which these models meet the identified requirements. The paper concludes by describing trends towards dynamic reference information models and addressing future challenges.

2. Towards mass-customisable ICT

This section discusses the characteristics of a mass customisation approach, and identifies the requirements when such an approach is applied to ICT.

2.1. What is mass customisation?

The term 'mass customisation' was coined by Davis [6]. It was initially promoted as a broad visionary concept, putting together seemingly contradictory notions: customisation to produce tailored products or services through flexible processes versus standardised mass production to produce in high volumes at reasonable costs [5,6]. Around the turn of the century, mass customisation was mentioned in the debate about agility versus leanness as a core concept to bridge the gap between both approaches. Agile strategies focus on flexibility and customisation, while lean strategies focus on efficiency and standardisation. From the debate it emerged that leanness and agility are not mutually exclusive strategies, and that a hybrid strategy of both agile and lean approaches is required to meet the requirements of responsiveness [10]. In such 'leagile' approaches, the positioning of Customer Order Decoupling Points (CODPs) plays a central role. The CODP separates that part of the supply chain geared towards directly satisfying customers' orders from that part of the supply chain anticipating future demand [11]. Downstream products are differentiated to specific customers or markets, while upstream products are standardised based on demand forecasts.

Based on the CODP position, a continuum of control strategies is proposed in the literature, varying from strategies in which all

processes are driven by customer order, to full anticipatory control in which all processes are based on demand forecasts. The main configurations are engineer-to-order (ETO), make-to-order (MTO), assemble-to-order (ATO) and make-to-stock (MTS). These strategies involve different levels of product differentiation, from fully customised in engineer-to-order (ETO) to fully standardised in make-to-stock (MTS).

Lampel and Mintzberg [12] argue that there is a dominant trend from both ends of the continuum towards the middle, namely towards mass customisation, or as they name it: 'customised standardisation'. Such a strategy offers the buyers the option of selecting their own set of components. Customers get their own product configuration but constrained by the range of available components. A mass customisation strategy is enabled by an assemble-to-order (ATO) production strategy, in which customer-specific products are assembled to customer order from standardised components that are fabricated to forecast.

Several authors stress that mass customisation imposes high demands on the enabling business systems. Pine et al. [5] argue that modularity and a linkage system are key factors to achieve mass customisation. Modularity allows parts of the product to be made in volume as standard modules, with product distinctiveness achieved through specific combinations of the modules [8]. A linkage system permits components to integrate rapidly in the best combinations or sequence required to tailor products or services [5]. Duray et al. [8] add that besides modularity, customer involvement in specifying the product is a key characteristic. Zipkin [13] states that mass customisation requires unique operational capabilities. He addresses four key capabilities: (i) a mechanism for interacting with the customer and obtaining specific information (elicitation); (ii) flexible processes that fabricate the product according to this information; (iii) the logistics to deliver the right product to the right customer; and (iv) powerful communications links that integrate these elements into a seamless whole. Tseng [14] identifies three basic requirements in design for mass customisation: (i) common building blocks that can be reused maximally, (ii) unified product architecture providing a structure of the defined building blocks that represent the capability of the enterprise to fulfil customers' needs in a unified manner; and (iii) a platform for meta level integration of the product realisation process. Huang et al. [15] focus on product platforms as a common success factor in achieving mass customisation, which encompass four strategies, i.e. commonality, modularity, scalability and postponement.

Synthesized from this literature, five key requirements for mass customisation systems are addressed:

1. *Generic product model*: mass customisation does not imply providing limitless choice, but it is restricted to available options. For quick product assembly, it must be clear what these possibilities are. Thus a product model is required that provides a standardised taxonomy representing different product variants and the underpinning structure [14]. Traditionally, product models are developed for every product or variant, which results in high complexity and component redundancy. Therefore, Hegge and Wortmann [16] introduced a generic product model that represents the set of all variants of a particular product family. It depicts all possible configuration options of product instances and the interdependencies that exist between components or features.
2. *Modularity*: products in a mass customisation approach consist of distinctive, autonomous and loosely coupled modules. These are independent components, each with its own single function and concentrated purpose [17]. These components can be used as black boxes, i.e. in order to use them, it is not necessary to know their inner structure, and changes in one component do

¹ The questionnaire can be obtained upon request from the corresponding author.

not impact the integrity of other components. Full product modularity also requires modularity of the enabling business processes and the network of producers and distributors.

3. *Integration platform*: modularity must go along with seamless integration in order to enable rapid and flexible assembly of customised products and services. Therefore, a platform is required that provides the production technology and communication infrastructure for flexible integration of product components. This platform assembles and interconnects the modules and coordinates the realisation of customised end products, according to given specifications including interface standards.
4. *Configuration support*: mass customisation is only possible if specific customer needs can be elicited rapidly and without error, while considering the possible options [13]. Furthermore, the identified requirements must be translated into the product information needed for procurement and manufacturing. This requires adequate systems for configuration, i.e. configurators. These are tools that guide users interactively through the product specification process. They show the available product range and provide possible selections based on the defined generic product model [18].
5. *Component availability*: in a mass customisation approach, products are assembled from an inventory of standard available components when specific customer orders appear. In order to meet short order-to-delivery lead-times, it is evident that the components with the right functionality must be available if customers demand a specific configuration.

2.2. Mass customisation of information systems

ICT mass customisation combines the seemingly contradictory notions of efficient standard software and flexible customised software. It enables customer-specific assembly of information systems from a repository of standard components. As such, mass-customisable ICT can be positioned in the middle of a continuum of standard packaged software and customised software. Software developers pre-design and realise modules based on forecasted functionality. Customers get their own ICT configuration, but constrained by the range of available components, as defined in reference models for the configuration of systems. These components could be supplied by different software vendors, which allows for using best-of-breed solutions in selecting and designing systems.

Following the identified requirements for mass customisation systems, the requirements for mass-customisable information systems are:

1. *Generic information model*: like product architectures in a mass customisation approach, information models should be set up as generic models, which define the class of architectures that can be assembled. Additional complexity of generic information models is that they comprise different interrelated model types including process, data and application models.
2. *Modular software*: modules in an ICT mass customisation approach are application-independent services, in which policy, input and output data, and interfaces are well defined (product modularity). They should not impose technical constraints on development of other modules (process modularity). Furthermore, it should be easy to replace a software module of provider A by a module of provider B, and it must be possible to combine modules of different vendors (network modularity).
3. *Information integration platform*: a software platform is required that the modules can easily be plugged into, that can enact the execution of modules upon the occurrence of external or internal events, and that enables the exchange of information

between the modules. Contrary to mass-customisable products, this platform has a virtual nature. It is not tied to one place. Especially internet-based techniques enable integration of modules that are located all over the world.

4. *Configuration support*: configuration of ICT elicits the required functionality of specific instantiations of information systems building upon a generic information model. Since information systems are composed of many interacting components, ICT configuration must be done for different levels of abstraction and different types of subsystems. Consequently, configuring information systems includes many partial configuration tasks that occur at different moments by different people. The dependencies between these different tasks must be well coordinated.
5. *Component availability*: the availability of software modules that, together, provide the desired functionality, including a specification of the interfaces. A specific characteristic of ICT components is again the virtual nature. This implies that components can be duplicated very quickly and at a negligible cost. On the other hand, availability is dependent on service providers, because users have access to the modules via an often complex information infrastructure.

During the last decades, much progress has been made in making ICT meet these requirements. Although it is not the purpose of this paper to analyse in detail technical implementations of ICT mass customisation, five important lines of developments can be distinguished: evolvement of modular software, shift from data-oriented to process-driven software, evolvement of Enterprise Application Integration (EAI), focus on architecture (model-driven software) and isolation of business logic in formalised rules. Service-Oriented Architecture (SOA) combines these advances into one integrated approach. SOA consists of three layers [19]:

- *A Business Process Management layer, coordinating the execution of business services*: this is a functional integration layer that groups services from the underlying business service layer into business processes. The process services are typically implemented through generic enactment engines, that execute workflows defined in languages such as BPEL or BPML. Following the workflow specifications, the enactment engines invoke services in the next layer. Services in the process layer can be rapidly configured or reconfigured using Business Process Management (BPM) tools.
- *A business services layer, delivering information services to the business processes*. The business services implement the information-processing functions of the actual business processes. Business services may be either straightforward data registration or reporting services, or complex services based on extensive business logic. They may implement these functions directly, for instance by applying business rules, or use application services that connect the business services to (legacy) information-processing application systems.
- *A business application layer, executing the application logic and data storage*. Applications are wrapped in application services, offering a standard web service interface to the business services, thus enabling Enterprise Application Integration (EAI).

The technological advances towards SOA have helped to meet, in particular, requirements b. and c. for mass-customisable information systems, i.e. software modularity and information integration platform. Moreover, it enables rapid system configuration by providing standards like Service Component Architecture that specify technical aspects of the interfacing and service assembly. However, SOA does not include the knowledge required

to specify services and to configure business processes as a sequence of services. Furthermore, the required software components must be available packaged as application-independent web services. Nonetheless, many current legacy systems are traditional monolithic ERP systems, which especially lack the required modularity [20,21].

So, even if a company applies SOA, important remaining challenges include the development of: (i) generic information models that specify families of business processes and services, and (ii) tools that support configuration of specific business process and service architectures. Reference models play an important role in meeting these requirements, as we will elaborate further in next section.

3. Role of reference models to enhance mass-customisable ICT

This section discusses the role of reference information models in mass-customisable ICT, and addresses the requirements to enhance such an approach.

3.1. What are reference information models?

Information modelling is an essential task in the analysis and development of information systems. Information models depict the architecture of mostly enterprise or supply chain systems. Architecture can be defined as the fundamental organisation of a system embodied in its components, their relationships to each other, and to the environment, and the principles guiding its design and evolution (IEEE 1471:2000). As such, architecture is about how specific products, systems or organisations are divided into subsystems and what the relationship is between these components (structural arrangement). Architectures of information systems are complex, since they consist of many different components and many different relationships. Information models are important in terms of making this complexity manageable, by providing systematic representations (visualisation, description) of architectures from different viewpoints and at various levels of abstraction. As such, information models support different stages of software development: requirements definition, design specification and implementation description.

Reference models have played an important role in information modelling for a long time, in particular in software engineering projects and implementation of standard packaged systems. For the purpose of this paper, reference information models should be distinguished from architectural frameworks. An architectural framework provides a systematic taxonomy of concepts of how to organise the structure of information models [22]. They depict the required information model types in different views and at various levels of abstraction, and show how these are related.

This paper deals with reference information models as conceptual models of information systems that formalise recommended practices for a certain class of domains [23]. Reference information models are used as a 'frame of reference' (i.e. blueprint, template) to construct company-specific information models and, as such, they improve quality, costs and lead-times of information modelling processes. Reference information models can contain different model types, depending on the purpose of usage specified in architectural frameworks. Most common are process and data models (including master data).

3.2. Continuum of reference information model strategies

Reference models are used in different ways in the ICT agility continuum, varying from pure standardisation to pure customisation. Below we describe the different use of reference models in these strategies, illustrated by a simplified example of configuring

a business process model (in BPMN notation) for the order fulfilment process of a discrete manufacturer that produces to customer order.

In a *pure standardisation approach*, standard packaged software is based on one standard reference information model, i.e. a 'dominant' design targeted to the broadest possible group of buyers. This standard model represents in detail forecasted functional requirements and accompanying characteristics of the system design. It is the starting point in software development. Implementing the software implies that the underlying standard reference information model is implemented, often implicitly. A company adopts the reference process model that is best suited to its characteristics. It is not possible to customise the reference model. Consequently, the instanced model is identical to the reference model.

Also, reference information models in a *segmented standardisation strategy* specify in detail the anticipated functional requirements as a basis to develop standard software. However, contrary to pure standardisation, there is not one dominant design, but reference models in this strategy cover multiple variants for different clusters of buyers. These reference models are used for selecting applicable functionalities and specification of the right parameters. A typical example of this strategy is ERP software that is segmented to different types of companies in different branches. Fig. 1 shows that, in this strategy, the reference model contains all allowed process variants present in this illustrative case: make-to-stock (MTS) and three types of make-to-order (MTO): discrete, continuous and batch production. Implementation implies selection of the most appropriate variant by skipping or blocking irrelevant options in the model. Subsequently, the instanced model is a stripped version of the reference model.

In a *customised standardisation (=mass customisation) strategy*, reference information models are generic models that specify the available components, possible features, and the permitted choices and combinations of components. The models are used both for design and development of software components and for guiding the configuration of customer-specific information systems. Fig. 2 shows that reference models in a mass customisation strategy depict generic building blocks that can be configured for specific situations. In this example, 'Production' is set up as a generic activity. All possible elements of this activity are defined, including the attributes and possible values. In this simplified example, two activities and two intermediate messages are available. Additionally, the reference model includes a tree of configuration choices and accompanying constraints. Configuration of the reference model implies configuration of specific instances by following a configuration tree. In this illustrative example, two choices are made. The production is order-driven, which implies that all elements are applicable. The production type is discrete, which implies that the activity Discrete Production is chosen. Based on the configuration choices, the values of the reference elements are specified. Thus, the instanced model is a specialised version of the reference model.

In a *tailored customisation strategy*, the starting point is a standard reference information model that specifies system design (just like in a pure standardisation strategy). At implementation, this reference model is modified for particular customer wishes. The customised reference model is used to develop a fully customer-specific software system, as far as possible reusing existing software components. Fig. 3 shows that in this approach companies adapt reference models until the customised model fully meets the user's specific requirements. In this example, adaptation of the reference model is achieved by inserting a missing activity (Plan Production), and changing an activity (Continuous Production into Discrete Production). Subsequently, the instanced model is a modified version of the reference model.

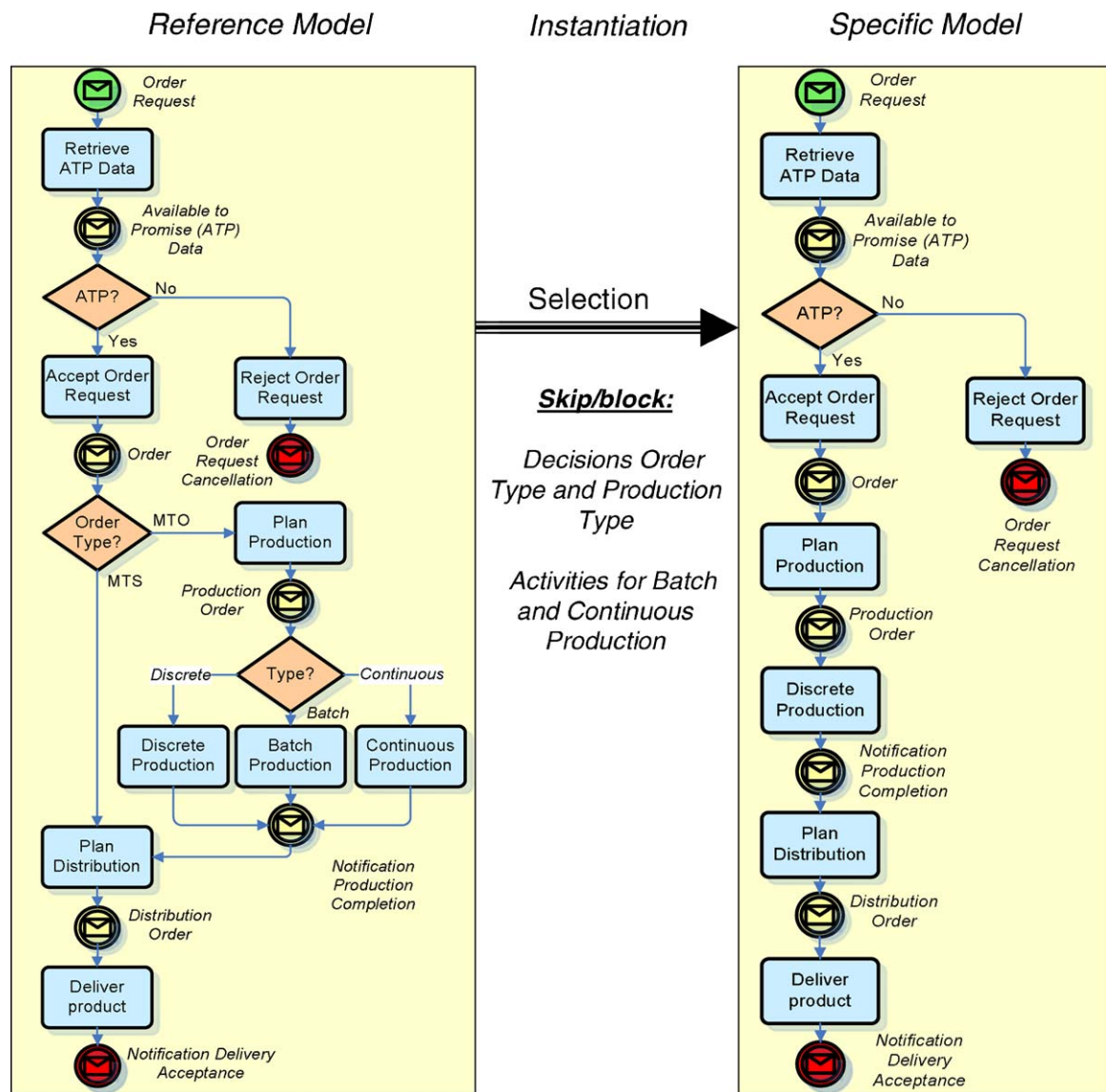


Fig. 1. Illustrative application of reference models in a segmented standardisation strategy.

Lastly, in *pure customisation*, strategy information systems are fully customised, without reusing existing designs and software modules. This implies that information modelling is done from scratch, fully according to customer specification, and reference information models are not used.

Overall, it can be concluded that reference models in these strategies mainly differ in the extent to which they can be customised to specific requirements. In pure standardisation, reference models are strict blueprints that must be taken as they are (one size fits all). In a segmented standardisation strategy, users can choose a specific variant that best suits their requirements, constrained by the available variants in the reference model. In a mass customisation strategy, users can specify their own configuration, but this is constrained by the available generic components and the rules that define permitted choices. In a tailored customisation strategy, users are free to modify the reference model to their specific requirements. However, it is important to notice that reference models are often used in a combination of strategies. For example: selection of the most appropriate variant from a segmented standardised reference model and then adaptation of this to the specific situation (tailored customisation). Also, different strategies can be applied at different abstraction levels.

In this paper, we focus on reference models used in an ICT mass customisation approach. Therefore, below we discuss further development of the characteristics demanded of such models.

3.3. Requirements of reference information models to enhance mass-customisable ICT

In accordance with the requirements for ICT mass customisation as developed in Section 2.2, the requirements for reference information models are:

1. *Generic model setup*: the most important requirement is that reference information models in a mass customisation strategy must be set up as generic models. They must specify the generic components of the supported information architectures, and define permitted choices and combinations in explicit configuration rules. Reference models must neither be single structures, as is the case in a pure standardisation and tailored customisation strategy, nor detailed specifications covering all possible variants, as is the case in a segmented standardisation strategy.
2. *Supporting software modularity*: in a mass customisation strategy, a reference model instance must be the design of a component-based information system. This requires that it

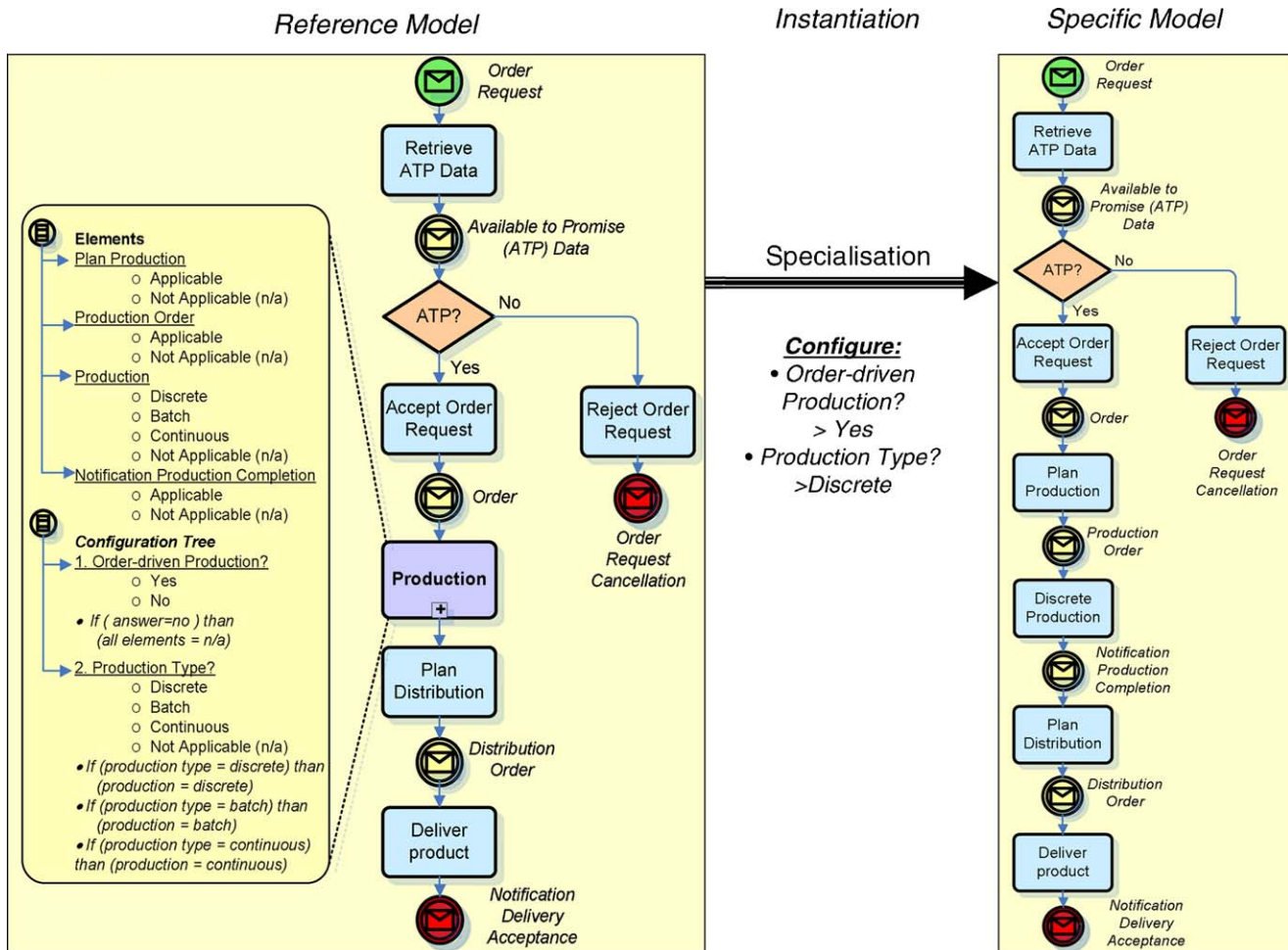


Fig. 2. Illustrative application of reference models in a mass customisation strategy.

provides precise definitions of the supported application services, including specifications of input and output data, policy, standardised interfaces, and possible sequences of services.

3. **Executable:** in mass customised systems, tailored functionality is provided by customised execution of standardised services. This is achieved by using the configured models in the run-time system for service orchestration, i.e. routing event data amongst multiple services. Therefore, the information models that support implementation must be machine-interpretable in the chosen software platform. In order to configure such models, reference information models must support a format that is executable. If reference information models are reused in different software environments, machine-interpretable requires that the reference model is represented in a standard notation.
4. **Configuration support:** the configuration of reference information models in a mass configuration approach is a complex task, especially in the domain of Production and Supply Chain Management. Therefore, there should be tools in place providing user-friendly means of leading users through the process of configuring specific information models. These tools must be able to inherit configuration choices, from requirements definition to implementation models.
5. **Content availability:** reference models must provide domain-specific knowledge required to specify services and to configure business processes as a sequence of services in specific areas of application. For example, dependent on the specific purpose of usage, in the domain of Production and Supply Chain Manage-

ment this might imply that the model includes: (i) both manufacturing, logistics, engineering, commercial and supporting processes including finance and quality management, and (ii) discrete, continuous and batch production, and (iii) different forms of forecast-driven and order-driven control (for example: material management with MRP, kanban and CONWIP), and (iv) both intra- and inter-enterprise integration.

4. Reference information models for Production and Supply Chain Management

Reference information models are important means to enable an ICT mass customisation approach, if they meet the requirements developed in the previous section. This section summarises the investigational results by providing an overview of existing reference information models and assessing the extent to which these models fit within a mass customisation approach.

4.1. Overview of existing reference information models

In the domain of Production and Supply Chain Management, widely acknowledged and applied reference models are SCOR, the GSCF Framework, VRM, the CPFR model of VICS, the MIT Process Handbook Y-CIM, ISA95, STEP, and the ERP reference models of SAP and Baan. Appendix A provides an introduction to these reference models. The models support different stages of information system development.

The MIT Process Handbook and the Supply Chain Management reference models, i.e. SCOR, VRM, and the GSCF framework,

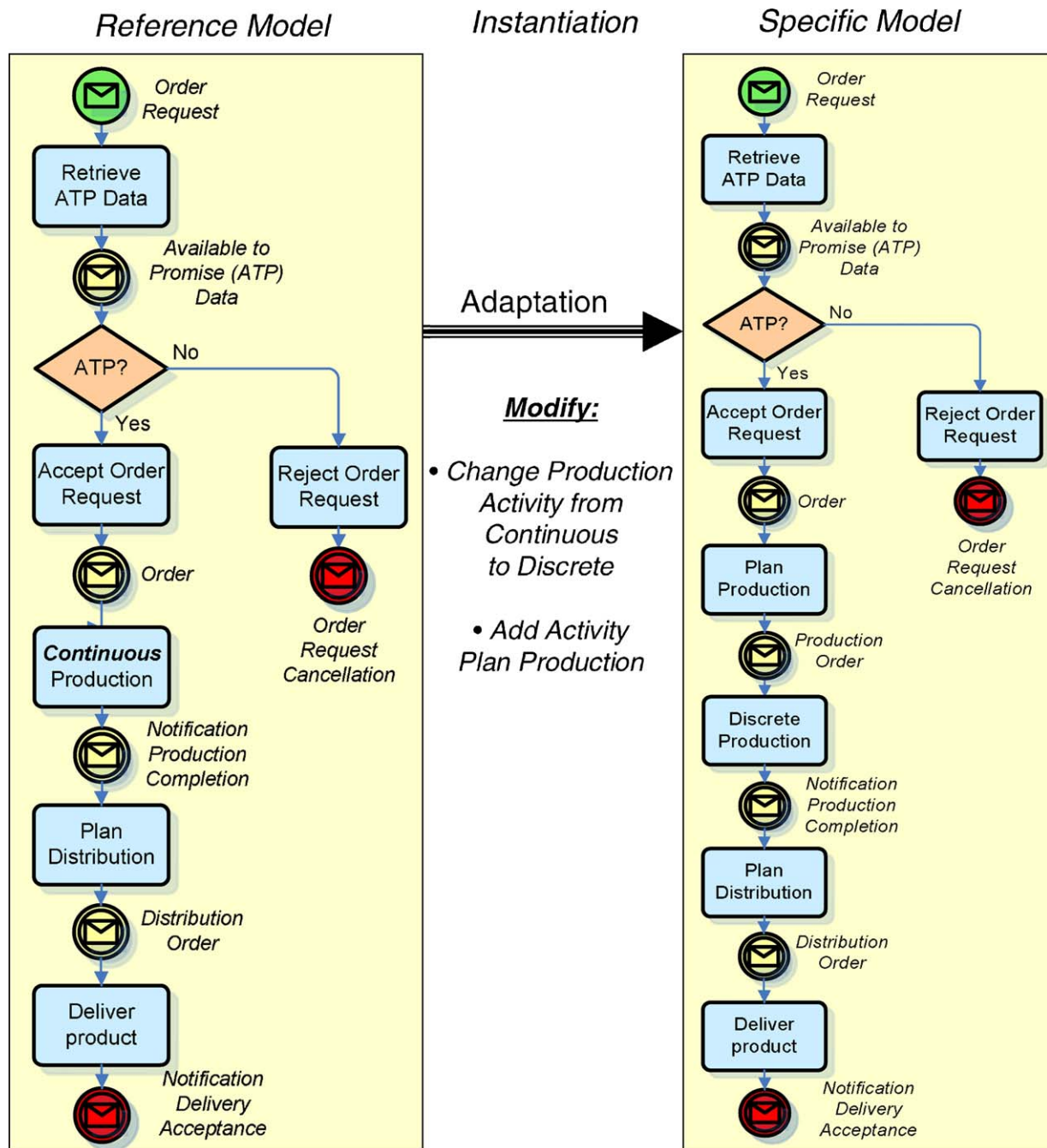


Fig. 3. Illustrative application of reference models in a tailored customisation strategy.

provide business knowledge that can be used to define information system requirements. The MIT Process Handbook focuses on a systematic repository of business knowledge of both single enterprise and supply chain processes [24]. The Supply Chain Operations Reference model (SCOR) aims to provide a reference model of supply chain processes [25]. Therefore, it includes process models of single enterprises (plan, source, make, deliver, return) and describes how these processes can be integrated in supply chains. The Value Reference Model (VRM, formerly VCOR) [26] and the framework of the Global Supply Chain Forum (GSCF) [27] have the same focus, although their scope is broader because they also support non-logistics processes such as marketing.

ISA95, Y-CIM, STEP and the CPFR model of VICS are focused on specification of functional design. ISA95 is a standard for integration of enterprise and manufacturing control systems

[28]. It includes interfaces between ERP and MES functions specified in XML. Y-CIM [29] aims at design of systems that integrate logistics, including production, and product engineering. STEP [30] is a standard including a reference data model for electronic exchange of product data. The CPFR model of VICS focuses on integration of retailers with manufacturers. It provides models for the collaborative aspects of planning, forecasting and replenishment processes and includes message specifications in XML [31].

The ERP models of SAP [32] and Baan [33,34] focus on implementation of their specific ERP software and, therefore, incorporate very detailed system-specific models. These models are mainly focused on single enterprises, and from that perspective they also support the interface processes, such as ordering, contracting, and integration of planning.

Table 1

Summarised overview of the extent to which existing reference models meet the identified requirement for ICT mass customisation.

Category	Reference model	Requirements				
		Generic model setup	Supporting software modularity	Executable	Configuration support	Content availability
Requirements	SCOR	–	n/a	n/a	–	+
	GSCF Framework	–	n/a	n/a	–	+
	VRM	–	n/a	n/a	–	+
Design	MIT Process Handbook	+/-	n/a	n/a	+/-	+
	CPFR model of VICS	–	+/-	+/-	–	+
	Y-CIM	–	+	+/-	–	+
	ISA95	–	+/-	+/-	–	+
	STEP	–	+/-	+/-	–	+
Implementation	SAP	+/-	+	+	+/-	+
	Baan	+/-	+	+	+/-	+

4.2. Usefulness of existing reference information models to enhance ICT mass customisation

Table 1 summarises the analysis of the extent to which the investigated reference information models meet the requirements for ICT mass customisation. It indicates that none of the investigated models sufficiently meet all of the defined requirements. Overall, in particular a generic model setup and adequate configuration support are missing. Most of the investigated models are based on a combination of (segmented or pure) standardisation and tailored customisation. An exception is the MIT Process Handbook, but this model lacks rules that define interrelationships between processes, and it focuses on textual knowledge that can be used in requirements definition. Furthermore, the ERP vendors, especially SAP, have performed valuable work in making their reference models configurable. Below, we elaborate further on the analysis of the different requirements.

4.2.1. Generic model setup

Most of the assessed reference information models are either single structures, as is the case in a pure standardisation and tailored customisation strategy; or detailed specifications covering all possible variants, as is the case in a segmented standardisation strategy.

An exception is the MIT Process Handbook, in which specialisation of processes is a core element [24,35]. In this handbook, processes are described as generic entities for which possible variants and attributes are defined. Companies that use the handbook to map their business processes, specify which variants are relevant to them, plus the company-specific values of the attributes. As such, the underlying concept fits well with a mass customisation approach. However, the handbook does not contain rules that specify interrelationships between processes. Furthermore, the focus is on providing a comprehensive repository of mainly textual business knowledge.

ERP vendors have incorporated elements of a mass customisation strategy in order to manage the enormous complexity of their detailed reference models covering all possible implementation variants.

In the Baan reference models, the detailed business process diagrams are linked to generic business functions by four possible types of rules [33]. Transformation rules specify which process is relevant for a certain business function. Static conditions can activate or de-activate parts of the linked processes. Parameter rules specify the settings of specific parameters. Consistency rules check the interdependencies between rules. However, this functionality has not often been used, mainly because users perceive it to be too complex.

SAP has conducted comprehensive research into the development, formalisation and testing of configurable reference modelling [36]. In this research, the EPC process modelling

language is extended to configurable EPCs (cEPC). cEPC includes a notation for configurable functions that can be blocked or skipped during configuration, and configurable connectors that can be made more specific during configuration. Moreover, interrelations within configurable models can be defined in the form of configuration requirements that constrain permitted choices, and configuration guidelines that define recommendations.

However, the reference models of SAP and Baan are still mainly based on a segmented standardisation strategy. They represent all possible implementation variants in detail. The configuration rules focus on selection of the appropriate variant by removing non-applicable variants and do not guide specialisation of generic models, as is the case in a mass customisation strategy.

4.2.2. Supporting software modularity

The assessed ERP reference models include process models that specify which components of their software are used in which activities. Furthermore, these models are integrated with SOA-based integration platforms, NetWeaver for SAP and Infor Open SOA for the Baan models. This is also the case for the Y-CIM model, since it is modelled in the ARIS toolset which is well integrated with NetWeaver and other SOA platforms.

STEP, ISA95 and CPFR/VICS have specified XML interface standards and thus these models partly enhance different applications to interoperate as modules. The MIT Process Handbook, SCOR, VRM, and the GSCF framework do not support software design and, thus, do not directly support software modularity.

4.2.3. Executable

The ERP reference models are well integrated with the applicable ERP systems. For example, Baan process models can be used both for dynamic generation of user-specific menus and for execution in the Baan workflow module. Furthermore, the models are compatible with SOA and, therefore, also executable by other systems.

Y-CIM, STEP, ISA95 and CPFR/VICS do not focus primarily on implementation. Subsequently they do not comprise executable models. However, STEP, ISA95 and CPFR/VICS include standard specifications in common computer-interpretable form (mainly XML). Furthermore, Y-CIM model is compatible with different ERP systems and SOA platforms, including NetWeaver, because of the use of the ARIS toolset.

The MIT Process Handbook, SCOR, VRM, and the GSCF framework only support business requirements definitions and, therefore, they are not executable.

4.2.4. Configuration support

The MIT Process Handbook offers users an internet tool to browse through the models and to zoom in from generic processes

to specialised variants. However, only the ERP reference models of both SAP and Baan provide functionality for guiding users through the model configuration process based on explicit configuration rules.

Building upon the SAP research into configurable EPCs (cEPC), Rosa et al. [37] propose a questionnaire-driven approach to guide a user interactively through the configuration of cEPCs by asking them questions, in natural language, and then linking the answers to the configuration choices defined in cEPCs. This approach is supported by tools for questionnaire specification and answering, definition of mappings between cEPCs and questionnaires, and generation of configured cEPCs. Together, these tools very nearly form a configurator for mass-customisable reference process models. However, an essential difference is that the input reference process models in cEPC notation are based on a segmented standardisation strategy that represents all possible variants in detail. This implies that the tools support selection of the appropriate variant and not specialisation from a generic model, as is the case in a customisation strategy.

In the Baan software, model wizards can be specified that guide users through selection of the right business process variant and through the setting of specific parameters [33]. These wizards can be added to business functions, and are based on the defined business rules. However, the Baan reference models do not contain predefined wizards. Furthermore, as discussed previously, the configuration rules were not often used, mainly because users perceived these to be too complex. Therefore, Baan experimented with the application of its software for product configuration to DEM process models, but due to the decline of the Baan Company at the end of the 1990s, this has not resulted in an operational system.

4.2.5. Content availability

The assessed models all have their own focus, but together they cover the main business processes for Production and Supply Chain Management. The ERP models focus on implementation of their specific ERP software and, thus, they provide very detailed and system-specific models. ISA95, CPFR and STEP focus on system-independent design of software interfaces, while Y-CIM aims at the design of platform-independent CIM systems. The MIT Process Handbook and the SCM models provide generic business knowledge that can be used to define information system requirements.

5. Outlook

5.1. Evolution of reference information models

If the investigated models are placed in a historical perspective, the following trend in development of reference information models can be distinguished as visualised, in a somewhat simplified manner, in Fig. 4.

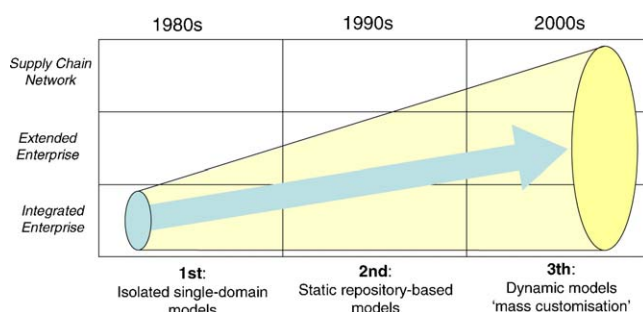


Fig. 4. Evolution of reference information models.

5.1.1. First generation: isolated single-domain models

Since the 1980s, reference information models have grown in popularity in the business community. There was a growing sense that software should be developed on the basis of architecture. Several architectural frameworks were developed. Subsequently, reference information models were developed. The emphasis was on isolated enterprise models in single domains, i.e. strict blueprints for specific industries (one size fits all). These models were used in the requirements definition and design phase of standard (standardisation strategy) or customised software (tailored customisation strategy). After having used the models intensively in setting-up software applications, the models were not integrated in the operational information systems. This resulted in a lack of active usage of the applied model and implementation experiences were mostly not incorporated into the reference models.

5.1.2. Second generation: static repository-based models

As from the mid-1990s, the standard software industry has developed reference models to accelerate implementation processes. Especially in the rising ERP industry, much effort was put into reference modelling, in particular by SAP and Baan. These models contained repositories of model variants that cover multiple domains (segmented standardisation strategy). During implementation, the relevant model variants were selected from the repository, and customised if necessary. Thereafter, the instantiated models were used for setting company-specific parameters and ultimately these were used in the run-time ERP systems. However, these second generation ERP reference models did not contain configuration logic. The rules for company-specific selection of model variants and parameter setting were implicit in the minds of implementation consultants.

In the late 1990s, eBusiness gained popularity, which put the emphasis on inter-enterprise processes. Enterprise reference models were expanded with architectures enabling integration with customers and suppliers, and these models started to become XML compliant. Furthermore, new models for supply chain management were developed, including the GSCF framework and SCOR.

5.1.3. Third-generation: dynamic models (mass customisation)

From the middle of the first decade of the century, there was a growing demand for mass-customisable information systems resulting in a renewed interest in reference models as a basis of information systems. Important driving forces included the new realism following the burst of the eBusiness bubble, which resulted in a strong emphasis on the returns from ICT, and the rise of the Service-Oriented Architecture (SOA) approach that combines technological progress made at different lines of ICT development. Also the ERP industry has embraced SOA and commenced to modularize their systems [38]. In addition, eBusiness developed beyond single B2C transactions, towards the support of different types of interactions in complex Supply Chain Networks (SCNs).

As a result of these developments, reference models are increasingly becoming compatible with SOA standards (such as BPEL) and made available in SOA-based integration platforms (such as NetWeaver for SAP). Moreover, ERP vendors, especially SAP, started to make their reference models configurable. On the other hand, in the SOA field the emphasis is shifting from technology towards business aspects, including architectural challenges such as business semantics for service definition, design methodologies for service engineering and business-oriented service composition [39,40]. Developing dynamic reference models that support a mass customisation strategy would be a valuable contribution towards accomplishing these developments.

5.2. Future challenges

The foremost remaining challenges for development towards third-generation models are setting up reference information models as generic models and incorporating configuration support. Existing models provide valuable knowledge, especially the progress accomplished by ERP suppliers to make their reference models configurable. However, current ERP reference models are complex because they represent all possible implementation variants in detail. Extending these models with explicit configuration logic will further increase the complexity of the reference models. Therefore, one of the key issues is to set up configurable reference models in a manageable and user-friendly way. This might be achieved by application of proven product and sales configurator software. At this, solutions must be found to deal with the redundancy of configuration logic in reference models and hard-coded in legacy systems.

Other important challenges are implementation-related. Despite the technological advances regarding SOA, systems today run on different technology and many current software applications are monolithic, non-modular systems. Replacing current systems with new flexible solutions is no realistic option for many companies, amongst others because of the significant investments in legacy and the risks of loosing stability. Therefore, an important precondition of dynamic reference models is the implementation of a SOA-based technical infrastructure that enables modularization and integration of existing information systems. Next, the incorporation of reference model configuration into systems execution will impact systems performance, in particular if the reference models are distributed amongst multiple organisations. In the latter case, also security will be an issue.

6. Conclusions

The objective of this paper was to assess how reference information models can be used to enhance ICT mass customisation. Therefore, it respectively aimed: (i) to identify the requirements to reference information models in an ICT mass customisation approach, (ii) to investigate the extent to which existing reference models, in the domain of Production and Supply Chain Management, are useful for ICT mass customisation, and (iii) to explore the trends in development towards dynamic models.

ICT mass customisation combines the seemingly contradictory notions of efficient standard software and flexible customised software. It assembles customer-specific information systems

from a repository of standard components. An important enabler is the availability of information models, which define the class of architectures that can be assembled. However, not all reference information models are appropriate for ICT mass customisation. Five requirements have been identified. The most important requirement is that they must be set up as generic models, which specify the generic components for configuration of specific information architectures, and that define permitted choices and combinations in explicit configuration rules. The reference models must support software modularity by precise definitions of the supported application services, including data specifications, interface standards and possible sequences of services. Furthermore, it must be possible to use the configured models in the run-time system for service orchestration. Also, the reference models have to include user-friendly means that guide users through the process of configuring specific information models. Lastly, the content reference models must provide domain-specific knowledge required in specific areas of application.

In an investigation of reference information models for Production and Supply Chain Management, it is found that none of the investigated models completely satisfy the defined requirements. Overall, particularly a generic model setup and sufficient configuration support are missing. Most of the investigated models are based on a combination of (segmented or pure) standardisation and tailored customisation. An exception is the MIT Process Handbook, but this model lacks rules that define interrelationships between processes, and focuses on textual knowledge that can be used in requirements definition. Furthermore, the ERP vendors, especially SAP, have done valuable work in making their reference models configurable. However, these models still represent all possible implementation variants in detail. Their configuration functionality is focused on selection of the appropriate variant and does not guide specialisation of generic models, as is the case in a mass customisation strategy.

When placing this analysis into a historical perspective, it emerges that reference information models have developed from isolated models for single enterprises, into static repository models for extended enterprises. At the moment there is growing demand for more dynamic reference information models, amongst other reasons because of the rise of the Service-Oriented Architecture (SOA) approach. Existing models provide valuable knowledge for that. The foremost remaining challenges are setting up reference information models as generic models, and developing user-friendly configuration support.

Appendix A. Appendix A. Reference models for Production and Supply Chain Management

Name	Stage	Introduction of the investigated reference models
SAP	Implementation	ERP market leader SAP developed detailed reference models to support SAP implementation processes [32]. Major elements of the model are process diagrams in the process modelling language Event-Driven Process Chains (EPCs) and data models. The EPC language has been conducted by SAP AG and the IDS Scheer AG in a collaborative research project in the years 1990–1992 [36]. The EPC models are integrated with the SAP ERP system and with NetWeaver, an SOA-based integration platform. The SAP reference model is set up for many industries. It contains different scenarios per major process. For example: the scenarios of production logistics are production by lot size, repetitive manufacturing, process manufacturing (including continuous production, discontinuous production and regulated production), 'make-to-order' production, and project-related 'engineer-to-order' [32].
Baan	Implementation	In the mid-1990s, the Baan Company developed reference models of 14 industries to accelerate and improve implementations. Major elements of these models are high-level Business Control Models (BCM), decompositions of business functions (including predefined parameter settings) and detailed process models (including roles, work instructions and links to specific Baan functionality). The reference models are developed with the tool DEM, which is a module of the Baan system [33]. User-specific menus can be generated from the implemented DEM reference models and the models can be used in the workflow module of the system. On the basis of the industry models, at the end of the 1990s, a multiple-domain reference model was developed, named the Hybrid Logistics Model [34]. This model is further developed in the Enterprise Business Model (EBM), which is currently maintained by Infor, the company that acquired Baan.

Appendix A (Continued)

Name	Stage	Introduction of the investigated reference models
Y-CIM	Design	The Y-CIM reference model was developed by Prof. Scheer in the early 1980s. Y-CIM encompasses integrated processes for logistics, including production, product engineering and the accompanying information and coordination processes [29]. The underlying modelling methodology was ARIS, which is further developed and commercialised in the ARIS Toolset of IDS Scheer AG, nowadays a leading supplier of business process modelling software. The Y-CIM reference model is broadly acknowledged in the industrial sector and has been developed into a comprehensive reference model that is also applied in other sectors.
STEP	Design	STEP (STandard for the Exchange of Product model data) is an ISO standard (ISO 10303) of which the first parts were published in 1994 [30]. It includes a reference data model for the electronic exchange of product data between computer-based product lifecycle systems, focusing on design and manufacturing applications. It covers a wide variety of different product types and lifecycle stages, such as design, analysis, planning, and manufacture. The information architecture of STEP is modelled in the Express language that can be represented in XML (STEP-XML, part of the standard).
ISA95	Design	ISA95 (formerly S95) is a standard of ISA for the integration of enterprise and manufacturing control systems, that is ERP and MES [28]. It involves functions for business planning & logistics, manufacturing operations & control, and control (batch, continuous, and discrete). S95 contains hierarchy models that describe the levels of functions and domains of control, a data flow model and an object model that focuses on the interfaces between MES en ERP systems, including XML specifications. The model builds upon The Purdue Reference Model, the MESA International Functional Model and the ISA S88 standard for batch control.
MIT Process Handbook	Requirements	The MIT Process Handbook was developed in the 1990s as a tool for (re)designing business processes and organising business process knowledge [24]. Important underlying concepts are the specialisation of processes based on inheritance notions and managing dependencies based on coordination theory. The handbook includes models of business activities, different business model archetypes, specific case examples, and frameworks for classifying activities. The activity models also include reference process models developed elsewhere, amongst others SCOR.
CPFR model of VICS	Design	In the short time since first publication in 1998, the Collaborative Planning, Forecasting, and Replenishment (CPFR) model of VICS has been acknowledged as best practice in business-to-business (B2B) commerce [31]. It provides a general framework for collaborative aspects of planning, forecasting and replenishment processes, focusing on the activities Strategy & Planning, Demand & Supply Management, Execution and Analysis. The framework contains process and data models. It addresses four different process variants: conventional order management, supplier-managed inventory, co-managed inventory and retail VMI. CPFR includes XML specifications that are integrated with the broader set of UN-CEFACT.
SCOR	Requirements	The Supply Chain Operations Reference model (SCOR) has been developed and endorsed by the Supply Chain Council (SCC) as the cross-industry standard for supply chain management [25]. The SCC was organised in 1996 by PRTM and AMR Research, and initially included 69 voluntary member companies (mainly in the U.S.). At the moment about 1000 companies are members of the SCC and the model is internationally acknowledged. The SCOR model focuses on the plan, source, make, deliver and return processes. It provides process models on three aggregation levels, standard process descriptions, performance metrics (in the categories Delivery Reliability, Responsiveness, Flexibility, Costs and Asset Management Efficiency), and best-practice descriptions.
VRM	Requirements	In April 2005 the Value-Chain Group was launched by former representatives of the Supply Chain Council. It has developed the Value Reference Model (VRM), which contains process models and metrics focusing on planning, governing and execution activities for product development, logistics and commercial processes [26]. The model is integrated with SOA standards including FERA.
GSCF Framework	Requirements	The Global Supply Chain Forum (GSCF) introduced another Supply Chain Model [27]. This model is built on eight key business processes that are both cross-firm and cross-functional, including functions such as production, R&D, logistics, marketing, purchasing and finance. The business processes that are addressed are Customer Relationship Management, Supplier Relationship Management, Customer Service Management, Order Fulfilment, Demand Management, Manufacturing Flow Management, Product Development and Commercialisation and Returns Management.

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